Third System
AI Presence Index

Three Empirical Regimes

AI Presence × Google Trends Across Three Categories

May 2026 · Pablo Ulpiano Gonzalez Castro, Third System

Independent measurement for the AI mediation layer.

The pre-registered prediction was that v0.11's PM software finding would generalise across categories. It didn't — and the falsification reveals something sharper.

PM software replicates v0.11 within 0.010 at both waves. Premium running shoes shows a structurally different regime: bivariate Spearman ρ = 0.81, but age and tier mediate 40% of the correlation; partial ρ ≈ 0.47. Premium olive oil routes to descriptive-only — 47% of the matched subset cannot be placed on the same scale: AI-present at 5-22%, but Trends below display threshold. **AI Availability is empirically separable from Mental Availability in three distinct ways.**

v0.11 established the AIAS construct-validity methodology in one category (project management software) and pre-announced a three-category expansion. v0.12 delivers that expansion: PM software is retained as the replication test, premium running shoes is added as a structurally different B2C category, and premium olive oil is added as a premium consumer packaged goods category with non-English production origins. The pre-registered prediction was that v0.11's boundary-mismatch signature would generalise across the three categories — the v0.11 marginal correlation magnitude, the leadership-zone divergence, and the Linear-style + Todoist-style diagnostic cases reproducing in 2 of 3 categories.

The prediction is falsified at both the H5 magnitude bar and the H6 diagnostic-case bar. The falsifications, however, organise into a sharper substantive finding: the empirical relationship between AI Availability and Mental Availability is **category-dependent**, with three distinct empirical regimes observable in the v0.12 sample. What unifies them is the persistence of category-boundary mismatch (H3 falsified in both confirmatory-eligible categories). What differentiates them is magnitude and structure.

Regime 1 — PM software replicates v0.11 with eerie precision. Spearman ρ = 0.506 at t_1, 0.482 at t_2 — within 0.010 of v0.11's 0.496 and 0.476 at both waves. H3 falsifies identically at 1 of 3 both waves. H6 confirms with Linear and Todoist surfacing as the same diagnostic cases. The v0.11 measurement is not a single-collection artefact — the construct correlation in PM software is structurally near 0.50 with concentrated boundary mismatch, reproducibly so across versions.

Regime 2 — Premium running shoes is a different empirical regime. Spearman ρ = 0.808 at t_1, 0.786 at t_2 — well above the moderate-to-strong threshold. H1 confirms decisively. But the partial ρ after age + tier control drops to 0.466 and 0.488 (a decrement of 0.34, against PM software's decrement of 0.07). Brand age and competitive tier mediate approximately 40% of the bivariate correlation. The strongest signal is driven by older incumbents (Asics, Nike, Adidas, Brooks, New Balance) showing jointly high AI Presence and high Trends. H3 still falsifies (2 of 3 both waves); H6 falsifies asymmetrically — Linear-style brands surface (Brooks at t_1, Hoka at t_2), but no Todoist-style brand surfaces in either wave.

Regime 3 — Premium olive oil reveals Category-Scale Mismatch. Routes to descriptive-only per pre-reg §3.4a (n = 8 at Worldwide, n = 7 at US, below the hard floor of 10). The substantive finding is sharper than the routing suggests: **7 of 15 matched-subset olive oil brands (46.7%)** have AI Presence rates of at least 5% at either wave, but Trends signal below the platform's display threshold. The two constructs cannot be placed on the same scale at all for nearly half the matched subset. The seven scale-mismatch brands include Frescobaldi Laudemio, Olio Verde, Lucini, McEvoy Ranch, Manni, Colonna, and Núñez de Prado — all returning the platform message 'Google Trends hasn't returned any results for this query' against the out-of-sample window.

The substantive contribution of v0.12 is the identification of these three regimes as distinct empirical manifestations of the same theoretical separability claim. The Tri-System Brand Growth framework's central claim is that AI Availability is empirically separable from Mental Availability — not equivalent at the level of practical inference. v0.11 supported this claim through a single-category boundary-mismatch finding. v0.12 supports the same claim more broadly: AI Availability is separable from Mental Availability in three distinct empirical ways across three structurally different categories. The framework's separability claim does not require a uniform mechanism of separation across categories; it requires that separability obtain. v0.12 demonstrates that it does, with the structural form of the separation varying with category characteristics.

What We Measured

v0.12 tests the cross-sectional correlation between two independent brand-level signals across three categories: per-brand AI Presence rate (drawn from v0.9's deposited matched-subset LLM measurements) and per-brand Google Trends search interest (acquired fresh under a pre-registered protocol). The design adds two cross-category hypotheses (H5 cross-category generalisation; H6 cross-category diagnostic-case detection) and a descriptive-only route for categories where the n-floor is breached, formalised in pre-reg §3.4a.

Three categories. Project management software (18 brands, retained from v0.11 as the replication test); premium running shoes (13 brands, new — a high-volume B2C category with well-known incumbents and DTC challengers); premium olive oil (15 brands in matched subset, new — a premium CPG category with non-English production origins). The three-category selection deliberately samples three positions on the cross-cutting dimensions of category volume, B2B vs B2C, and English vs multi-lingual production.

Per-category pivot rescaling. Each category has its own pivot brand (Asana for PM software, California Olive Ranch for olive oil, Asics for running shoes). Each pivot was validated for stability against the out-of-sample window prior to lock per pre-reg §6.A (California Olive Ranch mean 83.5, CV 12.4%; Asics mean 84.5, CV 7.9%). Pivot-rescaling converts each bundle-relative 0–100 daily value to a pivot-normalised daily value where the pivot is fixed at 100 by construction.

Two waves seven days apart. t_1 Trends window: 27 April – 3 May 2026. t_2 Trends window: 4 May – 10 May 2026. Both regions (Worldwide primary; US sensitivity). Single locked acquisition session at 2026-05-11T10:33:22Z covering 20 bundles × 2 regions = 280 (brand, day) cells.

Pre-registration. Locked at git commit ae4bd3a (tag *v0.12-prereg*) prior to any Google Trends acquisition call against the wave windows. Six per-category hypotheses (H1–H4 plus sensitivities), two cross-category hypotheses (H5, H6), and the Category-Scale Mismatch reporting rule (§10.2). Routing rule §3.4a: categories with fewer than 10 brands clearing E1b at either wave route to descriptive-only. The cross-category hypotheses become 2-of-2 effective if one category routes descriptive.

FINDING 01

PM software replicates v0.11 with high precision

Figure 1 · PM software rank-shift slope at t_1. Each brand's rank by AI Presence (left) connected to its rank by Trends (right). Steep slopes indicate construct divergence. Linear and Todoist surface as the v0.11 diagnostic cases at v0.12 — the same brands with the same structural positions.

v0.11's PM software finding reproduces under independent Trends acquisition seven days later. Spearman ρ = 0.506 at t_1 and 0.482 at t_2 — within 0.010 of v0.11's 0.496 and 0.476 at both waves. H3 falsifies identically at 1 of 3 both waves. The Linear paradox and Todoist inverse surface as the same diagnostic cases.

Spearman ρ at t_1 = 0.506 (n = 17; p_{1t} = 0.019); at t_2 = 0.482 (n = 18; p_{1t} = 0.022). The wave-to-wave direction of the just-miss flips: v0.11 missed at t_1 by 0.004, v0.12 misses at t_2 by 0.018. But the overall pattern reproduces within sampling error:

near 0.5 with the both-waves conjunction failing at the magnitude threshold.

H2 confirms with $|\Delta\rho|$ = 0.024 (v0.11 reported 0.020 — almost identical reproducibility across versions). The cross-wave stability of PM software's construct-validity correlation is itself reproducible. Whatever H1 measured at v0.11, it measured the same thing at v0.12.

H3 falsifies identically. At t_1, the AI-top-three is Linear, Asana, Notion; the Trends-top-five is Notion, Jira, Trello, ClickUp, Confluence. Only Notion overlaps. At t_2, the AI-top-three is Linear, Asana, Jira; the Trends-top-five is unchanged. Only Jira overlaps. Identical to v0.11 — 1 of 3 at both waves, same divergence concentration at the leadership zone.

H4 partial ρ = 0.417 at t_1, 0.434 at t_2, after age + tier control. Below the 0.5 magnitude bar at both waves — a decrement of 0.09 from bivariate. The covariate absorption is real but small. v0.11 reported 0.407 and 0.428; v0.12 is within 0.01 at both waves. Identical structural finding.

H6 diagnostic-case detection confirms. Linear is the unique Linear-style brand at both waves (AI Presence 86.5% at t_1, Trends rescaled 1.88). Todoist is the unique Todoist-style brand at t_1; Basecamp joins Todoist in the Todoist-style category at t_2. The v0.11 diagnostic cases are still the diagnostic cases at v0.12. PM software's construct-validity finding has direct test-retest validity at the category level.

FINDING 02

Running shoes is a structurally different regime

Figure 2 · Running shoes partial-correlation residual scatter at t_1. Rank residuals on both axes after OLS on rank-transformed brand age and tier ordinal. Bivariate ρ = 0.808 → partial ρ = 0.466 — brand age and tier mediate approximately 40% of the bivariate correlation.

Bivariate Spearman ρ = 0.808 at t_1, 0.786 at t_2 — well above the moderate-to-strong threshold and significant at $p_{1t} < 0.002$ at both waves. H1 confirms decisively. The partial ρ tells a different story: 0.466 and 0.488. Age and tier mediate 40% of the bivariate correlation.

H1 confirms; H4 falsifies; the decrement quantifies the regime. Bivariate ρ = 0.81 drops to partial ρ ≈ 0.47 after controlling for brand age and competitive tier — a decrement of 0.34. The corresponding decrement in v0.11 PM software was 0.09. Running shoes' strong bivariate signal is substantially confounded by joint variation in age-of-brand: older incumbents (Asics, Nike, Adidas, Brooks, New Balance) show jointly high AI Presence and high Trends, and that joint variation drives the correlation.

The residual partial correlation matches v0.11. Once age + tier is controlled, running shoes' direct AI-to-Trends construct correlation is approximately 0.47 — statistically indistinguishable from v0.11 PM software's partial of 0.42

(and from v0.12 PM software's 0.43). The categories are not actually different in the underlying relationship; they are different in how much of that relationship is explained by age-driven joint visibility versus direct construct correlation.

H3 falsifies at 2 of 3 both waves. At t_1, the AI-top-three is Asics, Brooks, New Balance; Brooks ranks sixth in Trends despite being top-three in AI Presence. At t_2, the AI-top-three is Asics, Brooks, Hoka; Brooks again falls outside the Trends top-five. The leadership-zone boundary mismatch is present, just less concentrated than PM software's.

H6 falsifies asymmetrically. Linear-style brands surface in running shoes (Brooks at t_1, AI Presence 56% with Trends rescaled 4.69; Hoka at t_2, AI Presence 51% with Trends rescaled 3.23). Both are performance-running incumbents with strong narratives in athletic discourse that LLM training data captures with high salience. But no Todoist-style brand surfaces in either wave — no running brand combines AI Presence below 5% with Trends rescaled above 20. The boundary mismatch is **unidirectional**: AI over-recommends certain incumbents, but does not under-recommend brands well-known to consumer search.

The substantive reading is that running shoes occupies a different empirical regime from v0.11 PM software despite sharing the boundary-mismatch mechanism. The bivariate correlation is strong; the magnitude is well above the v0.11 signature. But the direct construct-validity relationship after age and tier control is comparable to PM software's. The category's mechanism is not 'no boundary mismatch' but 'asymmetric boundary mismatch with age-driven bivariate covariance.'

FINDING 03

Olive oil reveals Category-Scale Mismatch

Figure 3 · Premium olive oil matched-subset bar visualisation. Horizontal bars showing per-brand AI Presence at t_1; green circles indicate brands with measurable Trends signal; copper × indicates brands with Trends signal below the platform's display threshold (EXCLUDED_E1a). The seven scale-mismatch cases are flagged ▯. Scale-mismatch index: 7 / 15 = 46.7% of the matched subset.

Premium olive oil routes to descriptive-only per pre-reg §3.4a (n = 8 at Worldwide; n = 7 at US; both below the hard floor of 10). The substantive finding is sharper than the routing suggests: **7 of 15 matched-subset olive oil brands (46.7%)** have AI Presence rates of at least 5% at either wave, but Trends signal below the platform's display threshold. The two constructs cannot be placed on the same scale at all.

Spearman ρ across the 8 PASS brands at Worldwide is 0.168 at t_1 and −0.095 at t_2 — essentially zero, with the wave-to-wave drift across the small PASS subset uninformative on its own. The PASS subset is too small for confirmatory inference; the substantive olive oil finding lives in the descriptive-only arm, not in the PASS subset correlation.

The seven scale-mismatch brands. Frescobaldi Laudemio (AI Presence 22–23% at both waves; Trends below threshold). Olio Verde (17–23%; below threshold). Lucini (18–20%; below threshold). McEvoy Ranch (5–9%; below threshold). Manni (3–5%; below threshold). Colonna (4–5%; below threshold). Núñez de Prado (4–5%; below threshold). All seven brands return the platform message 'Google Trends hasn't returned any results for this query' against the out-of-sample window at solo Phase B validation. The non-trivial AI Presence rates are not measurement noise — they are stable across waves and consistent across the two models in the matched subset.

The mechanism candidate is asymmetric specialist-vs-mass surface area. Premium artisanal olive oil brands have strong presence in food-publication discourse, restaurant recommendations, gift-guide listings, sommelier-style curation, and direct-to-consumer marketing channels that LLM training data captures. They have limited mass-market consumer-search interest under brand-name queries because the typical consumer of premium olive oil discovers brands through curation rather than direct search. The asymmetry is structurally different from the boundary mismatches in PM software (about category framing) and running shoes (about age-driven joint visibility); it is about the relative size of the specialist-recommendation surface versus the mass-search surface for the same brand.

Why this is the strongest separability evidence. In PM software and running shoes, AI Availability and Mental Availability are correlated but distinct (the v0.11 / v0.12 finding) or strongly correlated with shared age confound (the running shoes regime). In premium olive oil, the constructs are operating on different scales entirely: AI systems treat seven matched-subset brands as having meaningful category presence, while Google's consumer-search measurement infrastructure returns no signal at all for the same brand names at the same wave windows. The category-scale mismatch is the strongest empirical evidence in the v0.12 dataset that AI Availability is a separable construct — not just a tighter or differently-bounded version of Mental Availability, but a fundamentally different operational layer.

FINDING 04

Three regimes, one shared mechanism

Figure 4 · Cross-category H6 diagnostic zones at t_1. Linear-style region (top-left: AI ≥ 50%, Trends ≤ 5) and Todoist-style region (bottom-right: AI ≤ 5%, Trends ≥ 20) shaded in copper. PM software (left panel) shows both — Linear in Linear-style, Todoist in Todoist-style — replicating v0.11. Running shoes (right panel) shows Brooks in the Linear-style zone but no brand in the Todoist-style zone: the asymmetric boundary mismatch.

Across the three v0.12 categories, the shared mechanism is category-boundary mismatch (H3 falsified in both confirmatory-eligible categories). The differentiating dimensions are **magnitude of correlation** and **structural source of correlation**. Three distinct empirical manifestations of the same theoretical separability claim.

The shared mechanism. In PM software (1 of 3 top brands overlap), running shoes (2 of 3), and olive oil (constructs not on same scale entirely), AI's category boundary differs structurally from consumer search's. The specific brands surfaced inside AI's category are not the same brands surfaced inside consumer search's category. That boundary divergence is the shared mechanism.

Two differentiating dimensions. Magnitude: bivariate ρ ranges from near-zero in olive oil's PASS subset to marginal in PM software (≈0.50) to strong in running shoes (≈0.80). Structural source: direct construct relation (PM software, where partial ρ ≈0.42 is comparable in magnitude to bivariate ρ); age-mediated joint variation (running shoes, where bivariate ρ ≈0.80 drops to partial ρ ≈0.47); scale incommensurability (olive oil, where Trends signal is absent for nearly half the matched subset).

Why H5 / H6 falsifications are theoretically constructive. H5 (v0.11 signature generalises in 2 of 2 applicable categories) falsified for asymmetric reasons: PM software's t_1 ρ = 0.506 just clears the 0.5 threshold (within sampling error of v0.11's signature); running shoes is substantially above 0.5. The two falsifications carry different content. H6 falsified at 1 of 2 — PM software confirms with Linear and Todoist; running shoes lacks any Todoist-style brand. The asymmetric H6 falsification in running shoes is itself a substantive finding about that category's boundary structure.

The pooled rank-within-category sensitivity confirms the cross-category structure. Stacking rank-within-category data across PM and running (olive oil excluded for descriptive routing) yields Spearman $\rho = 0.624$ at t_1 ($n = 29$; $p < 0.001$) and 0.580 at t_2 ($n = 30$; $p < 0.001$). Within-category brand-ranking by AI Presence is positively associated with within-category brand-ranking by Trends, robustly across both confirmatory categories. The pooled correlation is stronger than v0.11's per-category ρ — driven largely by running shoes' strong bivariate signal.

The substantive contribution of v0.12 is the identification of these three regimes as distinct empirical manifestations of the same theoretical separability claim. The Tri-System Brand Growth framework's central claim — that AI Availability is empirically separable from Mental Availability — does not require a uniform mechanism of separation across categories. It requires that separability obtain. v0.12 demonstrates that it does, with the structural form of the separation varying with category characteristics.

Limitations

Single-acquisition design. The v0.12 Trends acquisition was executed at a single locked timestamp covering both waves. The replication evidence for v0.11 PM software is therefore between-version (v0.11 collection 10 May 2026; v0.12 collection 11 May 2026, one day apart) rather than within-version. A future v0.13 design with multiple independent acquisitions within the same pre-registered category would strengthen the test-retest validity claim at the methodology level.

Small olive oil n. The olive oil descriptive-only route is methodologically defensible per pre-reg §3.4a, but the n = 8 PASS subset is below the threshold for any meaningful correlation inference. The substantive olive oil interpretation rests on the absolute fact of scale mismatch — that 7 brands have AI Presence 5% with Trends below display threshold — rather than on any precise quantitative claim about the magnitude of the mismatch. A larger olive oil registry (50+ brands spanning the full premium-to-mass spectrum) would clarify whether the scale-mismatch index converges to a stable population value.

Bundle-padding methodology lesson. The olive oil Bundle 2 padding term (“extra virgin olive oil”) is a generic category-level search query with absolute search volume orders of magnitude higher than any individual brand in the bundle. This swamped the bundle's 0–100 relative-scaling reference at acquisition and produced quantization noise in the pivot-rescaled values for Brightland, Graza, and Kosterina. Documented in DEVIATIONS.md as a v0.13 design lesson: bundle padding must be brand-volume-comparable, not category-generic. The lesson does not modify any pre-registered hypothesis or routing rule and does not affect the categorical conclusion that olive oil routes to descriptive-only.

Single external validator. v0.12 tests construct validity against one external behavioural validator (Google Trends search interest). A multi-validator approach — testing AI Presence simultaneously against search interest, social-media mention rates, retail sales data where available, and consumer-survey aided-recall measures — would generalise the construct-validity claim from a single-validator finding to a multi-validator finding. Such a design is beyond the scope of v0.12 but is the medium-term goal of the AIAS programme.

Open question of practical predictive validity for purchase behaviour. v0.12 establishes construct validity at the category level for AI Presence against Google Trends search interest. The relationship between AI Presence and actual purchase behaviour at the brand level — the question that ultimately matters for the Tri-System Brand Growth framework's practical claims — remains the object of future Phase 3 work. The category-dependent nature of the AI-Presence-to-Trends relationship documented in this report suggests that the AI-Presence-to-purchase relationship is likely to be category-dependent as well.

What's Next

v0.13 — Full-category construct validity panel. Extends to five categories by adding premium facial skincare and personal finance to the v0.12 three-category set. Personal finance carries the Mint phantom-brand effect documented in v0.7 and v0.10 and is expected on theoretical grounds to show a **fourth empirical regime** distinct from the three documented here — phantom brands with non-zero AI Presence and exactly zero Trends signal sit at one tail of the construct-validity distribution. Premium facial skincare is structurally similar to premium olive oil (B2C consumer packaged goods with specialist-vs-mass surface asymmetry) and is expected to test whether the Category-Scale Mismatch regime reproduces in a structurally adjacent category.

Cross-lingual replication. v0.8's discourse-language bias finding documented that AI category boundaries depend on the language of the AI query. The construct-validity question in cross-lingual categories is whether the AI-Presence-to-Trends correlation depends on the language of the AI query, the language of the Trends acquisition, or both. Premium tea and traditional spirits are pre-identified candidate categories for the cross-lingual construct-validity test.

Tri-System Brand Growth manuscript (Gonzalez Castro, 2026, in preparation). The v0.12 three-regimes finding sharpens the framework's empirical content. The category-dependent nature of the AI Availability / Mental Availability relationship is a candidate empirical anchor for the framework's separability claim — AI Availability as a third system whose relationship to Mental Availability varies systematically with category structure.

Causal-mechanism studies. v0.12 establishes covariation across three regimes; the next-step question is what specifically about each category produces its regime. Brand age, category-discourse surface asymmetry, and category-framing tightness are candidate mechanisms in the three regimes documented here. Mechanism studies operationalising those candidates are the medium-term research path.

Phase 4 (AIAS components 2–6). Construct validity established for AI Presence in v0.11–v0.13 is a precondition for measurement work on Ranking, Consistency, Coverage, Grounding, and Sentiment. The v0.12 three-regimes finding suggests that the construct-validity bar for each AIAS component is unlikely to be a clean uniform correlation with any single external proxy across all categories.

Hypothesis Scoring

All thresholds and tests locked at *v0.12-prereg* (commit ae4bd3a, 11 May 2026 UTC) prior to any Google Trends acquisition call against the wave windows. Per-category pivot exemption from E1b applied per pre-reg §5.1 (sd = 0 by pivot construction). Categories routing to descriptive-only per §3.4a (n < 10 hard floor) carry no H1–H4 inference.

H	Pre-registered prediction	Result	Status
H1 PM	Spearman $\rho > 0.5$ AND $p_{1t} < 0.05$, both waves	$\rho_{t_1} = 0.506$; $\rho_{t_2} = 0.482$	FALSIFIED
H2 PM	$\lvert\Delta\rho\rvert$ [illegible] 0.15	$\lvert\Delta\rho\rvert = 0.024$	**CONFIRMED**
H3 PM	AI top-3 Trends top-5, both waves	1 / 3 at both waves	FALSIFIED
H4 PM	Partial $\rho > 0.5$, both waves	partial $\rho_{t_1} = 0.417$; partial $\rho_{t_2} = 0.434$	FALSIFIED
H1 Run	Spearman $\rho > 0.5$ AND $p_{1t} < 0.05$, both waves	$\rho_{t_1} = 0.808$ ($p < 0.001$); $\rho_{t_2} = 0.786$ ($p = 0.001$)	**CONFIRMED**
H2 Run	$\lvert\Delta\rho\rvert$ [illegible] 0.15	$\lvert\Delta\rho\rvert = 0.022$	**CONFIRMED**
H3 Run	AI top-3 Trends top-5, both waves	2 / 3 at both waves	FALSIFIED
H4 Run	Partial $\rho > 0.5$, both waves	partial $\rho_{t_1} = 0.466$; partial $\rho_{t_2} = 0.488$	FALSIFIED
H1–H4 Oil	n-floor [illegible] 10 per wave	n = 8 (WW), n = 7 (US); below hard floor	*Descriptive-only per §3.4a*
§10.2 Oil	Category-Scale Mismatch index	7 / 15 = 46.7% of matched subset (AI [illegible] 5% AND Trends below display threshold)	*Descriptive finding*
H5	v0.11 signature in 2 of 2 applicable categories	0 of 2 (PM ρ just clears 0.5 at t_1; Running ρ above 0.5)	FALSIFIED
H6	Linear-style + Todoist-style in 2 of 2 applicable categories	1 of 2 (PM confirms; Running missing Todoist-style)	FALSIFIED

Hypothesis Details

Per-hypothesis claim, operationalisation, and result. The both-waves conjunction for H1 / H4 provides built-in family-wise error rate control (joint $p \approx 0.05^2 = 0.0025$ under the null). Cross-category H5 / H6 evaluate against the effective 2-of-2 rule because olive oil routes to descriptive-only.

PM software replication. v0.11's PM finding reproduces under independent Trends acquisition seven days later. H1 falsifies at the magnitude bar at one wave (t_1 = 0.506 just clears 0.5; t_2 = 0.482 below). H2 confirms with $\lvert\Delta\rvert = 0.024$. H3 falsifies identically at 1/3 both waves with Notion / Jira as the overlapping brands. H4 partial = 0.42 (t_1) and 0.43 (t_2) — covariate absorption is real but small (decrement of 0.09 from bivariate). The construct in PM software is reproducibly near the marginal threshold with concentrated boundary mismatch.

Premium running shoes. H1 confirms decisively (bivariate ≈ 0.80 both waves, $p < 0.002$). H2 confirms $\lvert\Delta\rvert = 0.022$. H3 falsifies at 2/3 both waves (Brooks AI-top-3 but Trends rank 6). H4 falsifies with partial ≈ 0.47 — a decrement of 0.34 from bivariate, indicating that age + tier mediate approximately 40% of the bivariate correlation. The residual partial correlation matches v0.11 PM software's partial ≈ 0.42, suggesting the direct AI-to-Trends construct relationship is comparable across categories once the age-of-brand confound is removed.

Premium olive oil. Routes to descriptive-only per §3.4a (n = 8 Worldwide, n = 7 US; both below the hard floor of 10). The Category-Scale Mismatch finding (§10.2): 7 of 15 matched-subset brands have AI Presence [illegible] 5% at either wave but Trends signal below the platform's display threshold. The seven scale-mismatch brands (Frescobaldi Laudemio, Olio

Verde, Lucini, McEvoy Ranch, Manni, Colonna, Núñez de Prado) constitute the strongest separability evidence in the v0.12 dataset: the two constructs cannot be placed on the same scale at all for nearly half the matched subset.

H5 cross-category generalisation. v0.11 signature defined as < 0.5 AND $p < 0.05$ AND H3 falsified, at both waves. Applicable categories: PM software and running shoes (olive oil descriptive). Required: 2 of 2 effective. **FALSIFIED.** PM software fails because t_1 = 0.506 just clears the 0.5 threshold (within sampling error of v0.11's signature; the wave-to-wave direction of the just-miss flipped). Running shoes fails because is in a different magnitude regime entirely (≈0.80 — well above the v0.11 signature). The two falsifications carry different substantive content.

H6 cross-category diagnostic-case detection. Linear-style brand (AI 50% AND Trends 5) AND Todoist-style brand (AI 5% AND Trends 20) surface at both waves in 2 of 2 applicable categories. **FALSIFIED at 1 of 2.** PM software confirms: Linear-style = Linear at both waves; Todoist-style = Todoist (and Basecamp at t_2). Running shoes confirms Linear-style (Brooks at t_1, Hoka at t_2) but no Todoist-style brand surfaces. The asymmetric H6 falsification in running shoes is itself a substantive finding about that category's unidirectional boundary mismatch.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.1). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

Gonzalez Castro, P. U. (2026). ***Three Empirical Regimes: AI Presence × Google Trends across PM Software, Running Shoes, and Olive Oil — AI Presence Index v0.12 Phase 3 expansion***. Third System. thirdsystem.ai/v12-three-empirical-regimes (SSRN 6748341)

Underlying datasets

OSF project ec6wh, /v12/. Inputs (v0.9 AI Presence rates), Google Trends raw responses (20 pivot-bundle JSONs across 2 regions — PM software 5 bundles, olive oil 2 bundles, running shoes 3 bundles), topic-ID suggestion logs, pre-acquisition validation outputs (solo and bundled E1a checks), brand age source table (46 rows across 3 categories), registry files frozen at v0.6's final state, scoring outputs, build scripts, this report, and the matching SSRN working paper.

Companion SSRN working paper (Gonzalez Castro 2026, SSRN 6748341). Cross-references: AI Availability foundational paper (SSRN 6659000); AIAS Presence Measurement Protocol v1.1 (SSRN 6722319); v0.6 Cross-Category Findings (SSRN 6720959); v0.7 Phantom-Brand Persistence Phase 2 BBB (SSRN 6721779); v0.8 Discourse-Language Knives (SSRN 6728000); v0.9 Longitudinal Re-Baseline (SSRN 6736878); v0.10 Naive-Phantom Rate Stability (SSRN 6741163); v0.11 PM Software × Trends Construct Validity Pilot (SSRN 6745040).

Methodology log: v0.12 follows AIAS Presence Measurement Protocol v1.1 (unchanged from v0.9). Pre-registration locked at git tag v0.12-prereg (commit ae4bd3a) on 11 May 2026 UTC prior to acquisition. Acquisition session UTC timestamp 2026-05-11T10:33:22Z. DEVIATIONS.md Entry 1 documents the olive oil Bundle 2 padding methodology lesson as a non-design-altering observation; no pre-registered hypothesis, threshold, or routing rule was modified..

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